Report

Insecure Data Storage & Sensitive Data Exposure

Suyash A Gunjal

Insecure Data Storage :

nsecure Data Storage essentially refers to data packets or data that is stored without the added protection of encryption or other firewalls. It can also refer to unintentional data leaks caused by improper application design.

This data can be accessed maliciously in two ways when it comes to mobile applications:

1. If the threat agent has access to the physical device itself and
2. Using Malware to gain access to sensitive data remotely.

Risk factors

Insecure data storage can arise from:

- Storing sensitive data in plaintext without encrypting it.
- Saving sensitive data in external storage (SD cards or similar) without protection, which is accessible to other apps or physical attackers.
- Failing to enforce strict file permissions, allowing unauthorized apps or users to access stored data.
- Sensitive data cached in memory or local storage that is not protected by encryption or cleared after use.
- Storing encryption keys alongside the encrypted data or failing to protect keys adequately.
- Using weak or predictable encryption keys.

Insecure Data Storage: Threats & Impacts

As stated earlier, access to sensitive data is the end goal of any hacker. This is the worst-case scenario for users and can leave organisations open to serious reputational risk. The maliciously obtained data can also lead to the following:

- Identity Theft
- Fraud

- Privacy violations
- External Policy Violations (PCI)
- Material Loss

Sensitive Data Exposure:

Sensitive data is anything that should not be accessible to unauthorized access, known as sensitive data. Sensitive data may include personally identifiable information (PII), such as Social Security numbers, financial information, or login credentials. Sensitive Data Exposure occurs when an organization unknowingly exposes sensitive data or when a security incident leads to the accidental or unlawful destruction, loss, alteration, or unauthorized disclosure of, or access to sensitive data. Such Data exposure may occur as a result of inadequate protection of a database, misconfigurations when bringing up new instances of datastores, inappropriate usage of data systems, and more.

- Confidentiality Breach: where there is unauthorized or accidental disclosure of, or access to, sensitive data.
- Integrity Breach: where there is an unauthorized or accidental alteration of sensitive data.
- Availability Breach: where there is an unauthorized or accidental loss of access to, or destruction of, sensitive data. This will include both the permanent and temporary loss of sensitive data.

Sensitive Data Exposure - Threats & Impacts

Identity Theft:

Attackers can misuse personal info (name, DOB, Aadhaar/PAN, etc.) to impersonate the victim.

Account Takeover (ATO):

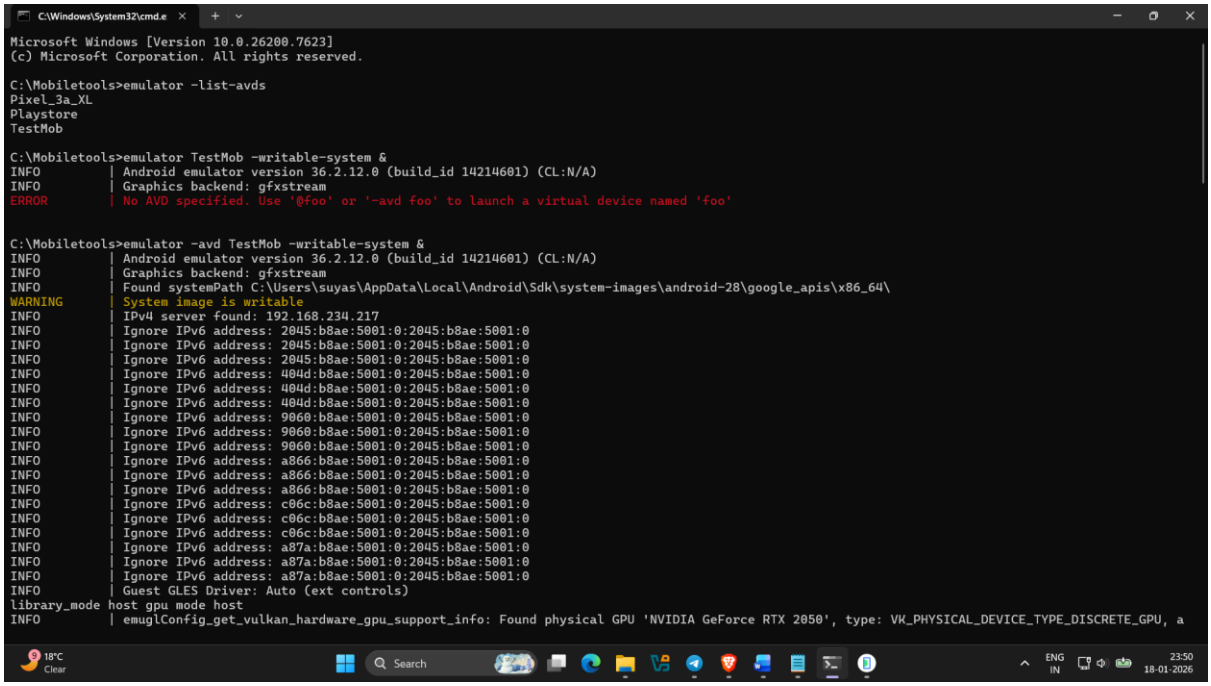
Leaked passwords, OTPs, tokens, or reset links allow attackers to log in and control accounts.

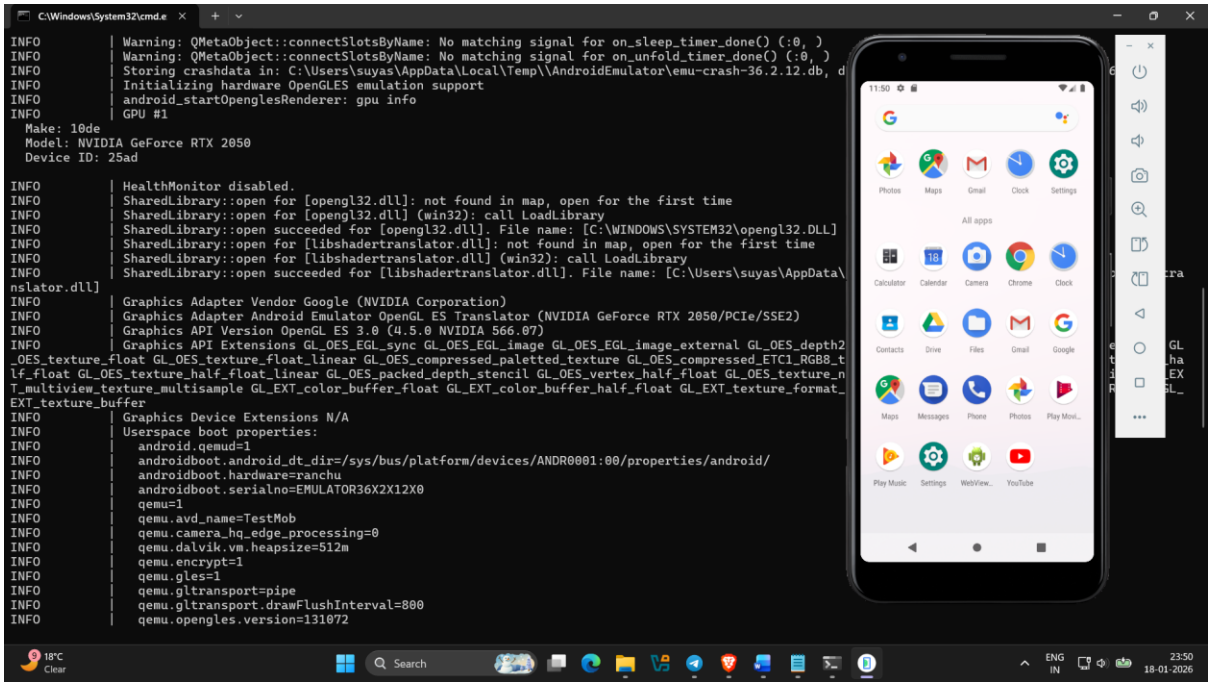
Financial Fraud:

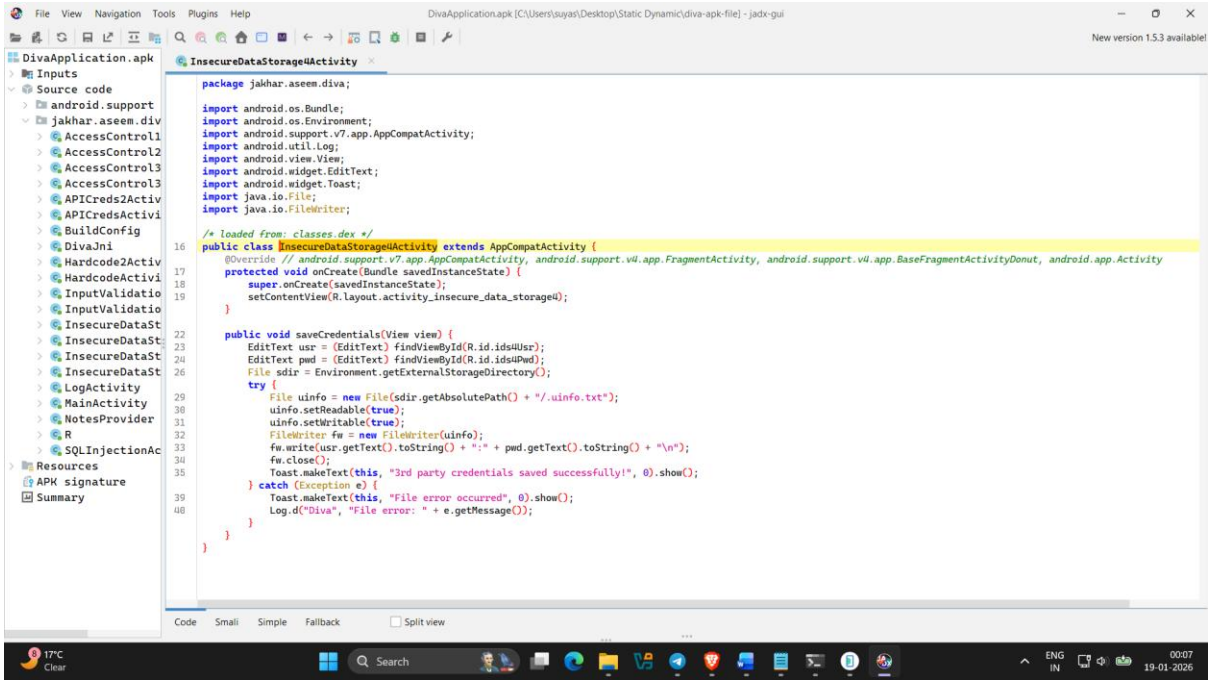
Exposure of bank details, card info, or UPI data can lead to unauthorized transactions.

Session Hijacking:

If session cookies/JWT tokens leak, attackers can access accounts without login credentials.







References:

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